

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

An Earth observation that survives an agent handoff.

EMEM gives an observation a lookup identity and a content-addressed record. Agent A passes the reference. Agent B resolves the same record and checks it before continuing.

Earth data stays in the data layer. Its reference enters the reasoning layer.

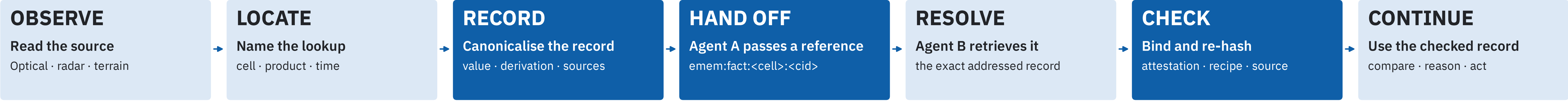
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Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortx-AI/emem (Apache-2.0)

1 How does Earth data enter an agent workflow?

Keep the source data upstream. Pass its reference.

Locate an observation, record what was read, and carry its content address into the next agent's reasoning.

EARTH DATA



Lookup identity: where / product / time

Content address: BLAKE3 of the canonical record

CONTROLLED HANDOFF TEST: ACTED ON CORRUPTED EVIDENCE

prose
254 / 276

JSON
164 / 276

retrieved text
220 / 276

opaque id
154 / 288

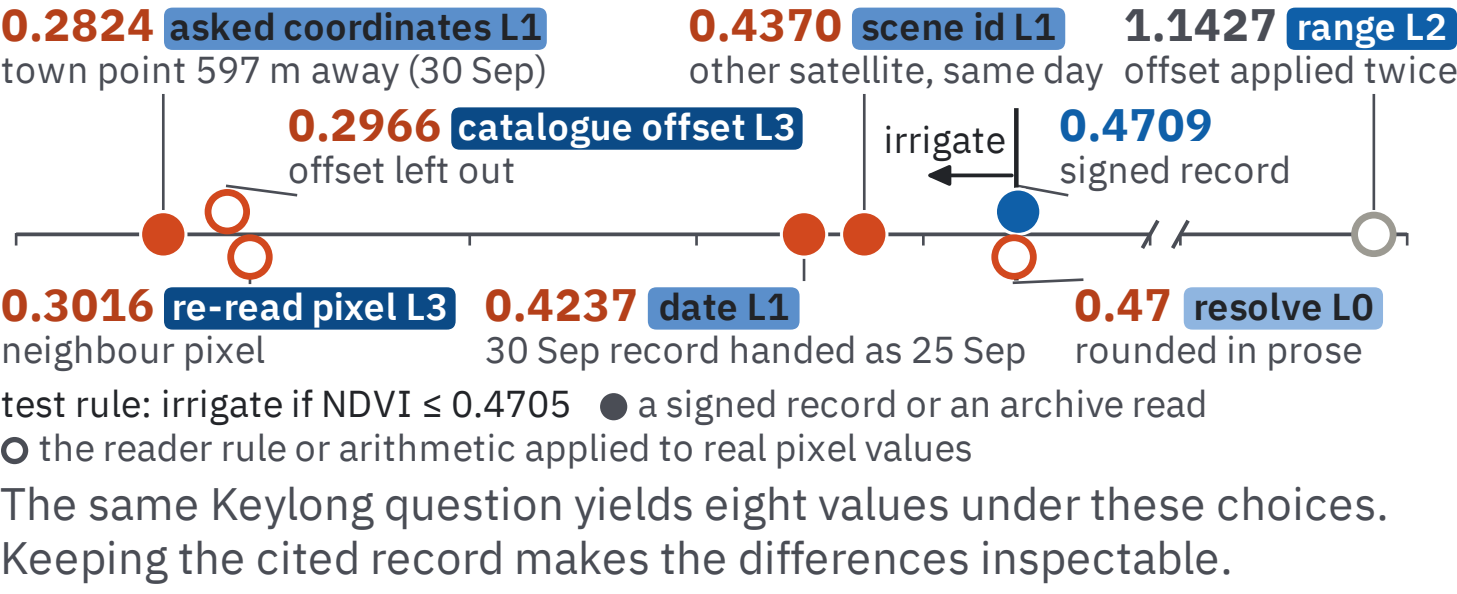
checked reference
0 / 300

In the controlled handoff test, prose led agents to act on 254 of 276 corruptions; a reference with explicit checking instructions, 0 of 300.

2 What can change between agents?

One question, several records

Scene, pixel, offset, rounding and stale state can change the value passed on.



3 Where did the evidence change?

A changed value needs a cause

Keep the scene, pixel, model and time attached so a change can be investigated.

$$\Delta Z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \epsilon$$

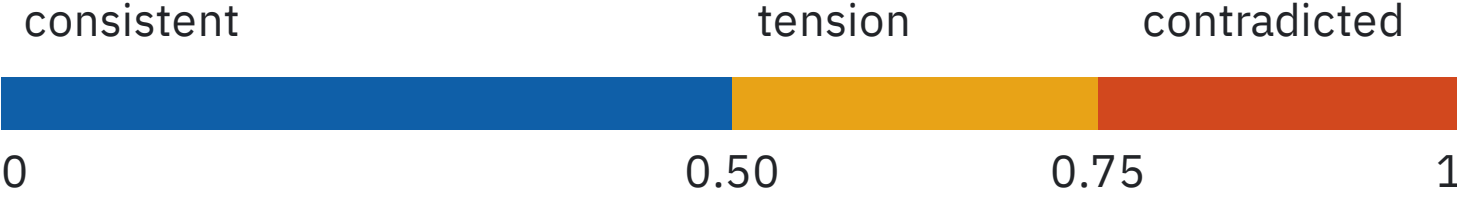
World, instrument, alignment, model and noise. This is an attribution model; the numeric split remains open work.

IMPLEMENTED · DRIFT-ANCHOR SCORE

$$u = |x - a| / (3\sigma)$$

$$s = u / (1 + u)$$

x: output · a: anchor · σ: anchor uncertainty



Tests pin 3σ → 0.50 and 9σ → 0.75.

Missing or invalid α: exact agreement required.

The implemented anchor score measures disagreement, not its cause. SAT-042 exercises it in a reference harness; live device-to-anchor wiring remains next work.

Memory = observations + edges

$$M = (O^*, E^*)$$

O* stores observations; E* stores typed temporal relations. Append new records and keep earlier ones.

$$E = (\text{subject}, \text{predicate}, \text{object}, t_{\text{from}}, t_{\text{to}})$$

Relations can say supersedes or disagrees_with, with a validity interval.

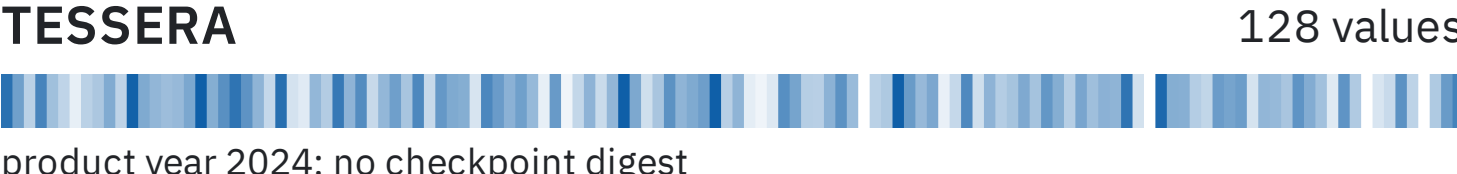
Lookup: (cell, band, observation time)
cid(O) = BLAKE3(CBOR_{canonical}(O))

recall(M, a, b | t*, τ)

At address a and band b, bound observation time by t* and what memory knew by signing time τ.

Model outputs are records too

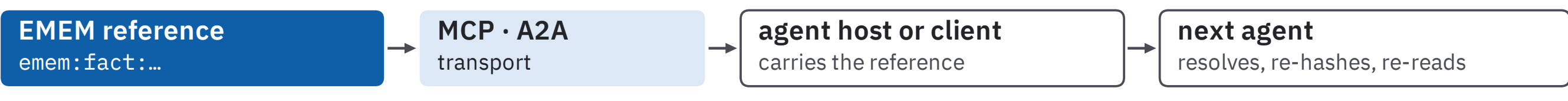
Satellite input → encoder → vector record → content address



Archived vectors: Prithvi names a checkpoint digest; TESSERA names a product year. These encoders are retired on the deployment; stored records remain addressable.

11 The same reference can cross agent runtimes

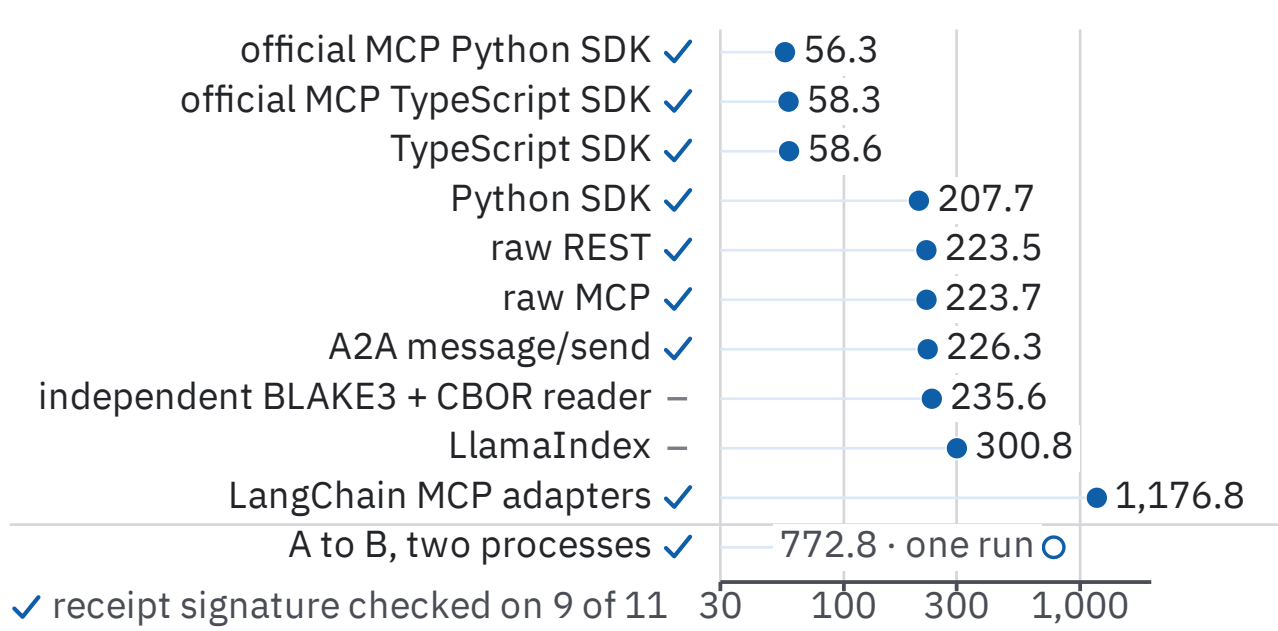
MCP, A2A, REST and SDKs expose the same reference model. One reference returned the same record and value through 11 client paths; receipt signatures were checked on 9 (30 Sep 2026). Surface status is labelled below.



AGENT HOSTS	INTEROPERABILITY	DISCOVERY	DEVELOPER	FRAMEWORKS
● Claude Code ● Claude Code plugin ● Claude.ai custom connector ● Dify Marketplace plugin (community)	○ Model Context Protocol (MCP) emem.dev/mcp ○ Claude2Agent (A2A) signed agent card	□ Official MCP Registry io.github.Vortx-AI/emem □ GitHub MCP Registry ● github.com/Vortx-AI/emem □ Glama	● Gemini CLI ● Visual Studio Code ○ Cursor ● pip install ememdev ● npm i @vortxai/emem ○ REST / OpenAPI 3.1 ● ghcr.io/vortx-ai/emem	△ LlamaIndex △ AutoGen △ CrewAI △ Mastra

● LIVE connected or published ○ PROTOCOL exposed interface □ REGISTRY a listing △ EXAMPLE repository adapter
checked 30 Sep 2026 ChatGPT (@emem) · PUBLISHER LISTING

one address, one value on every path
median ms of 3 runs, log scale, 30 Sep 2026



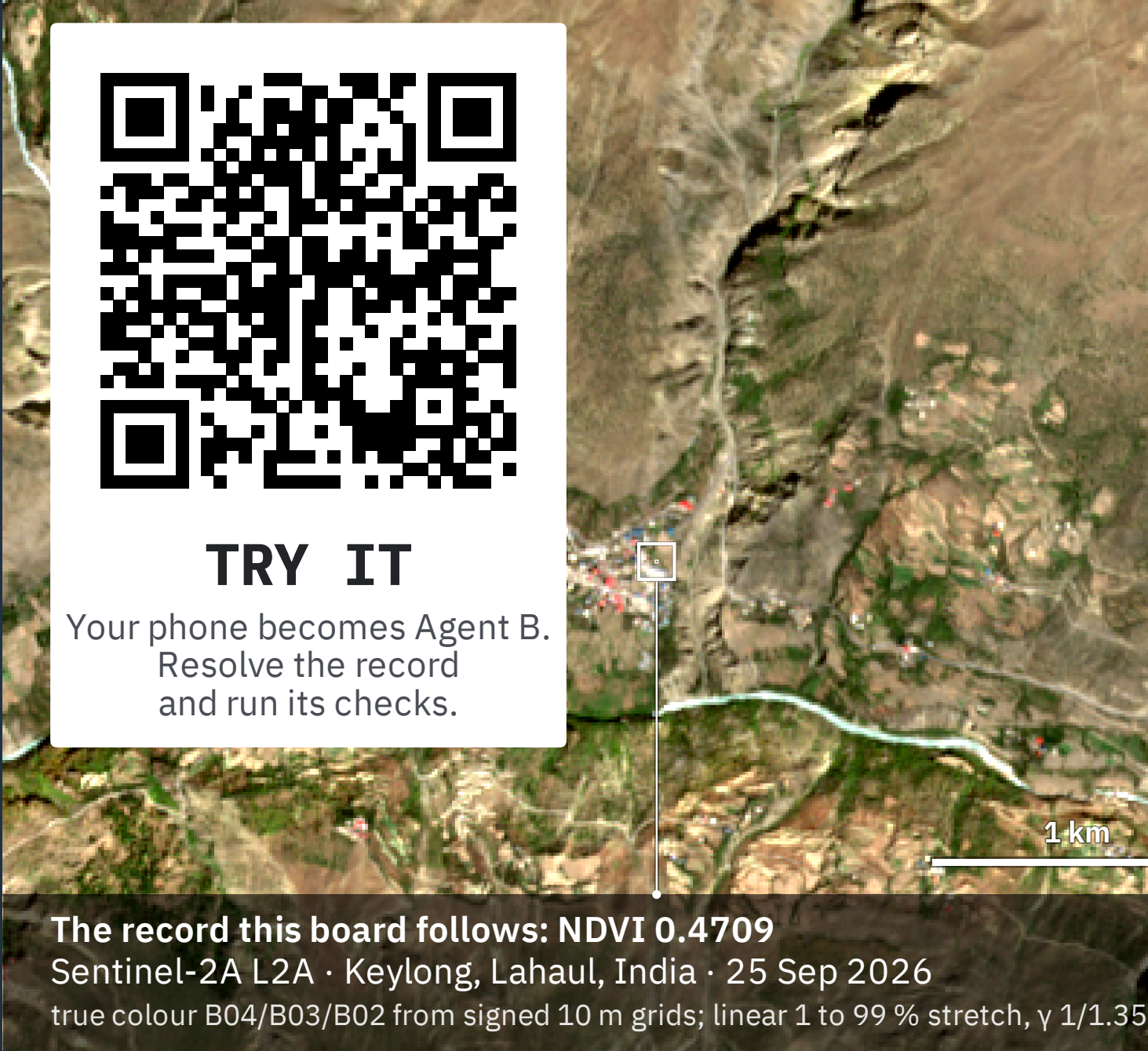
12 EO data into agent reasoning

STAC describes assets; openEO defines processing; PROV and C2PA carry provenance. EMEM adds the observation reference an agent can pass on and resolve again.

JUDGE	GeoGuard	a claim in text	resolve · re-hash · re-read
CARRY	MCP, A2A	a tool call, an agent card	hand off
RETRIEVE	RAG	a text chunk	
SIGN FILES	C2PA, CDSE Traceability	a file	
RECORD LINEAGE	W3C PROV	entity, activity, agent	
RUN	openEO	a process graph	
FIND	STAC	an asset (file)	

An observation stays addressable across a handoff. The next agent can recover the cited record, inspect its provenance, recompute a declared recipe and return to its source.

REPRODUCE
Records, code, experiments and integrations.
vortx-ai.github.io/esa_poster



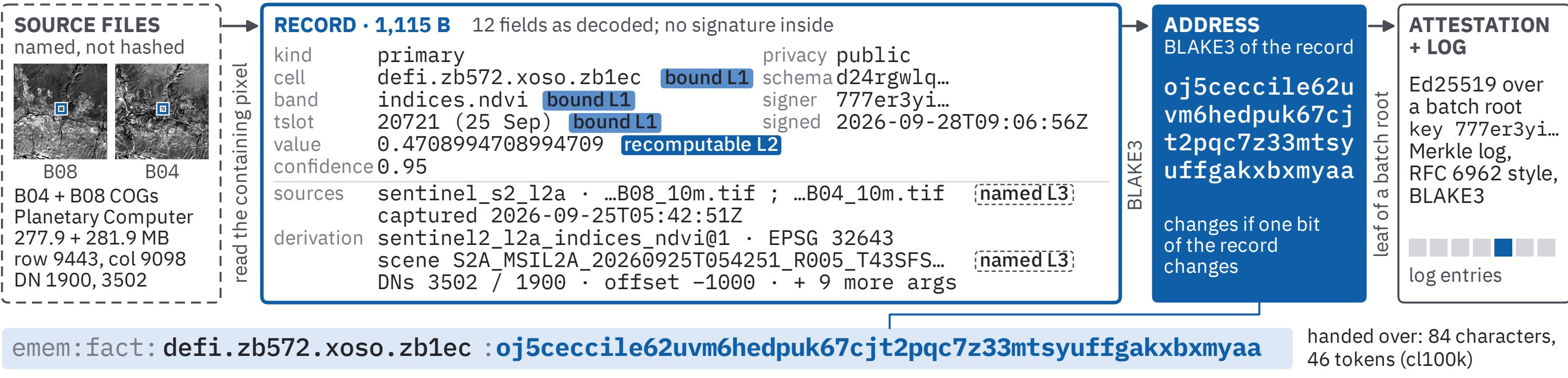
The record this board follows: NDVI 0.4709
Sentinel-2A L2A · Keylong, Lahaul, India · 25 Sep 2026
true colour B04/B03/B02 from signed 10 m grids; linear 1 to 99 % stretch, y 1/1.35

4 What exactly is handed over?

A compact reference opens the observation record

Resolve the value, place, time, derivation and source references from the record's address.

INSPECT The 1,115 bytes behind this example.



BLAKE3 identifies the exact 1,115-byte record. A batch attestation covers its address. The record names the source files and read location; source checking continues by re-reading them.

5 Which changes can the receiver detect?

Checks can continue back to the source

Resolve → Re-hash → Bind → Recompute → Re-read

agents as B: three Claude models pooled, 1 Oct 2026	prose	JSON	RAG	opaque id	EMEM, full check	first check that refuses
■ B acted on corrupted evidence ■ M2 seen in production	■ unaffected ■ refused ■ not applicable	■	■	■	■	■
G0 nothing altered: accepted everywhere, never refused						
value M1 stated value moved by 1 ULP	12/12	12/12	12/12	9/12		C L0
M2 stated value rounded to "0.47"	6/12	6/12	11/12	0/12		C L0
M8 value 0.45, re-encoded and re-hashed	12/12	6/12	12/12	6/12	F	F L0
cell M4 record cited for another cell	7/12	0/12	6/12	6/12	E	E L1
M9 cell changed inside the record, re-hashed	7/12	11/12	6/12	6/12	F	F L0
time M10 tslot changed to look current, re-hashed	12/12	6/12	6/12	6/12	F	F L0
band M6 a record for another band handed over	12/12	0/12	6/12	2/12	E	E L1
source M11 source scene id changed, re-hashed	12/12	12/12	12/12	12/12	F	F L0
derivation M12 offset changed, value recomputed, re-hashed	12/12	11/12	12/12	8/12	F	F L0
M14 value (0.46) disagrees with the signed DNs	12/12	6/12	12/12	6/12	H	H L2
stale / current M5 an older record handed over as current	12/12	1/12	6/12	1/12	E	E L1
M16 a second signed version shown only to B	12/12	6/12	6/12	6/12	G	G L0
signature / id M7 1 ULP changed inside the served bytes	12/12	12/12	12/12	12/12	D	D L0
M3 token miscopied by one character	n/a	n/a	n/a	0/12	C	C L0
M13 value 0.45 under the forger's own key	12/12	6/12	12/12	6/12	F	F L0
source pixel M15 DNs read from the pixel 10 m south	12/12	12/12	12/12	12/12	I	I L3
entity M17 same record; A meant another entity (out of scope)	not run	not run	not run	not run		none

B acts on corrupted evidence

C resolve L0 D hash L0 E binding L1 F signature L0 G log L0 H re-read L3 D to I are cumulative: each adds one check
Bars: agents' false acceptance, k of n per cell, n 12 (6 Haiku 4.5, 5 Sonnet 5.5, 1 Opus 5.5 runs); squares: the deterministic ceiling. Totals pool 23 items, 1 Oct 2026.
Three further signer errors (offset, same-day scene, unit) pass checks D to I; metadata checks catch them.

Each step answers a different question. The matrix measures which changes become visible at each depth; signatures and log checks establish attestation and publication history.

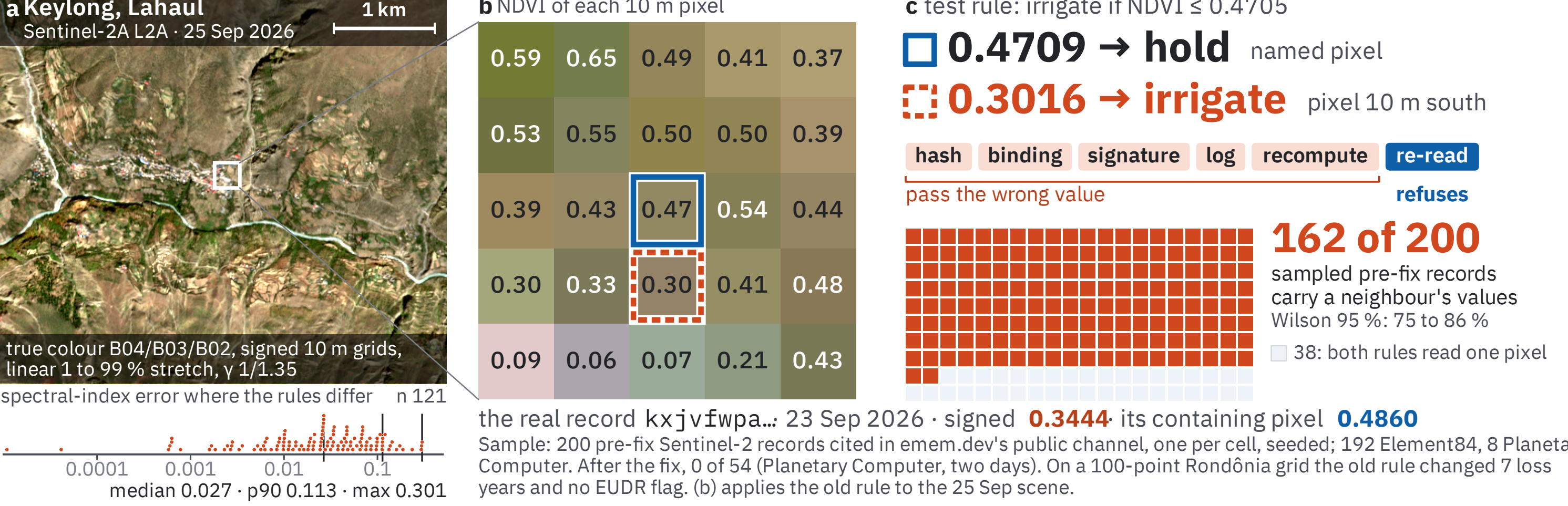
Agent bars: 12 runs per item, across 23 items. JSON and an opaque id caught some changes from their fields. In the live pre-fix record case, agents acted on 4 of 4 uninstructed, 2 of 4 with resolve-and-bind instructions, and 0 of 4 with a source re-read.

6 What does returning to the source add?

A valid record can preserve a wrong source read

$$NDVI = (DN_8 - DN_4) / (DN_8 + DN_4 + 2\sigma), \sigma = -1000$$

The recorded product offset σ is applied to both bands before the ratio.



A record from the earlier reader still resolves: 0.3444 recorded, 0.4860 at the containing pixel. The source re-read exposes the pixel-selection difference.

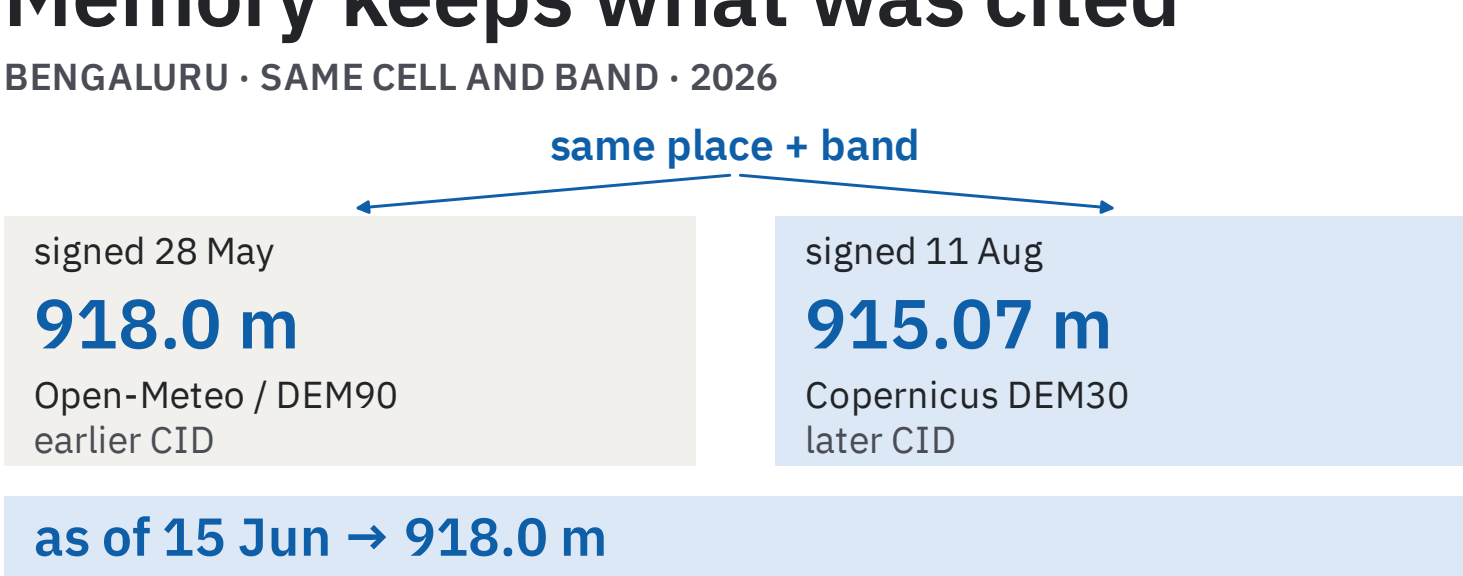
7 What has the receiver established?

Know what was checked

RECORD	Exact addressed bytes
BINDING	Requested place, time and product
ATTESTATION	Signature under the pinned key
DERIVATION	Declared computation reproduces
SOURCE READ	Named source supports the value

Source quality remains inherited. Entity meaning and downstream decisions belong to the application using the evidence.

8 Can a later agent recover the earlier state? Memory keeps what was cited

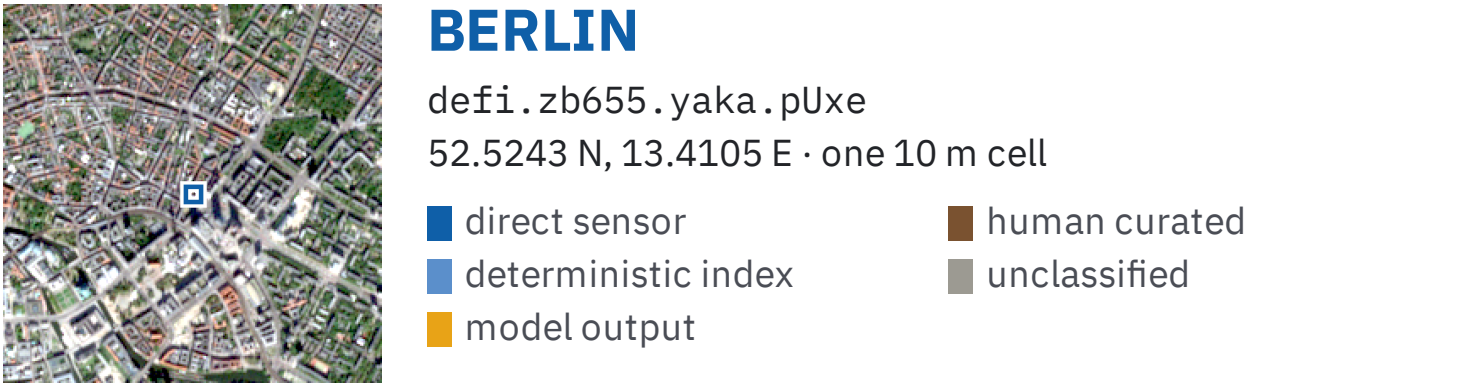


The next agent can recover the earlier citation. Asked what memory knew on 15 Jun, recall returns 918.0 m. The later 915.07 m record reflects a provider change; it does not establish ground movement.

9 What does one Berlin location reveal?

One place, many observations

Central Berlin: optical, radar, elevation, temperature, water, forest and biomass. One lookup cell organises 15 products, preserving each record and native grid.



product	signed reading	valid time	grid	fact_cid
Sentinel-2C L2A	NIR 0.0475	27 Sep 2026	10 m	wtdq64...
Sentinel-2C L2A	NDVI 0.195	27 Sep 2026	10 m	y2k2ti...
Sentinel-1C RTC	VV -2.82 dB	28 Sep 2026	n/s	2x2en...
GLO-30 DEM	36.6 m DSM	2021	30 m	sdu3gf...
MODIS MOD11A2	297.4 K day LST	29 Aug 2026	1 km	yqhihm...
ESA WorldCover	built-up (50)	2021	10 m	xicg6u...
ESA CCI Biomass	0 t/ha	2022	100 m	keebyc...
JRC GSW V1.4	0 % water	1984 to 2021	30 m	hohgux...
Hansen V1.13	0 % tree cover	2000	n/s	x4kpkj...
JRC GFC2020 V4	not forest	2020	10 m	oog2b4...
Copernicus CAMS	NO ₂ 12.3 µg/m ³	30 Sep 2026	~11 km	jvwhwg...
Overture, not EO	0 buildings	Aug 2026	vector	czehax...
ISRIC SoilGrids	no data, null	30 Sep 2026	250 m	mkwinv...
CHIRPS v2.0	outside ±30°	30 Sep 2026	0.05°	bkaovt...
JRC TME	outside ±30°	30 Sep 2026	30 m	w7t2hx...
NASA FIRMS	no fire in 24 h	30 Sep 2026	points	k2ykb2...

15 products, 16 signed facts, 4 signed absences, 10 m to about 11 km; read live 30 Sep 2026. Native grids remain distinct; n/s: grid not in source files. 16 records re-hash under emem.dev's key, 2 Oct 2026.

10 What else can an agent carry?

Beyond a single observation

OBSERVATIONS	fact · derived value · absence
Recover a value and its derivation.	
EARTH DATA	raster · cube · rasterset
Recover a field or a series of fields.	
COLLECTIONS	bundle · tree
Group facts or address file chunks.	
IDENTITY AND PROCESS	entity · state · trace · track
Name an object, a reasoning stage or a run.	

Choose the reference that preserves what the next agent needs to recover.

NEXT EXPERIMENTS

Measure each cause of change

Validate numeric attribution under controlled scene, pixel and encoder changes. Repeat checked handoffs between separately operated agent hosts.